

Knocked Unconscious: Thematic Change in Popular Hip Hop Songs

Deborah Olorunisola

Yale University

Yale

Research Questions



- RQ1: What are the most prevalent themes in hip hop?
 - Rap music is known for flagrant discussions of explicit themes—sexuality, drugs, violence, and more. [2]
- RQ2: Has rap music become less socially-conscious over the years?
 - “Non-puterile” rap (introspective and/or socially-conscious) is said to be on the decline. [1]

Data

The first source is the Billboard “Hot Rap Songs” chart from March 11th, 1989 until December 31st, 2024 with 4,598 unique songs. This data is available via the `billboard.py` API. The second source is *Ego Trip's Book of Rap Lists*, which supplements the missing lyrics data between 1979 to 1988 with 397 unique songs. Songs from these reference books are available via Genius.com. Song lyrics for all the songs ($n = 4,994$) come from Genius.com via the `LyricsGenius` API.

Methods

Multilabel Classification

Define a dataset $\mathcal{X}$ with points $\mathbf{x} \in \mathbb{R}^d$. The goal is to find a mapping $\mathcal{X} \rightarrow \mathcal{Y}$ for l labels. This mapping is unknown, so we use machine-learning to train a classifier using labeled training data $\mathcal{D} = \{(\mathbf{x}_1, \mathbf{Y}_1), \dots, (\mathbf{x}_N, \mathbf{Y}_N)\}$, where $\mathbf{Y}_i$ is the specific set of labels (one-hot encoding) corresponding to observation $\mathbf{x}_i$. The text embeddings representing each $\mathbf{x}$ were from a pre-trained BERT model (`Lyrics-BERT`).

Classifiers Employed

1. Binary Relevance (BR)
2. Label Powerset (LP)
3. Classifier Chain (CC) with imprecise correlation ordering
4. Multilabel K-Nearest Neighbors (MIKNN)
5. LLaMa 3.2 (LLM)

Key Findings

1. There is little difference in the distribution of rap themes over the years (Fig. 1, 3, 4).
2. MLC methods picked up on thematic spikes aligning with the Bling and Don Eras of hip hop, suggesting their utility for classification on short text in high dimensions (Fig. 2).
3. The genre appears heavily associated with braggadocio and street culture. (Fig. 3, 4)
4. According to both models, socially conscious and introspective rap have never made up *most* of the popular songs in an era. (Fig. 1)

Results

Figure 1. Differences in the prevalence of puerile rap, according to the top-performing models. Non-puterile rap is defined as any song with themes of social-consciousness or introspection.

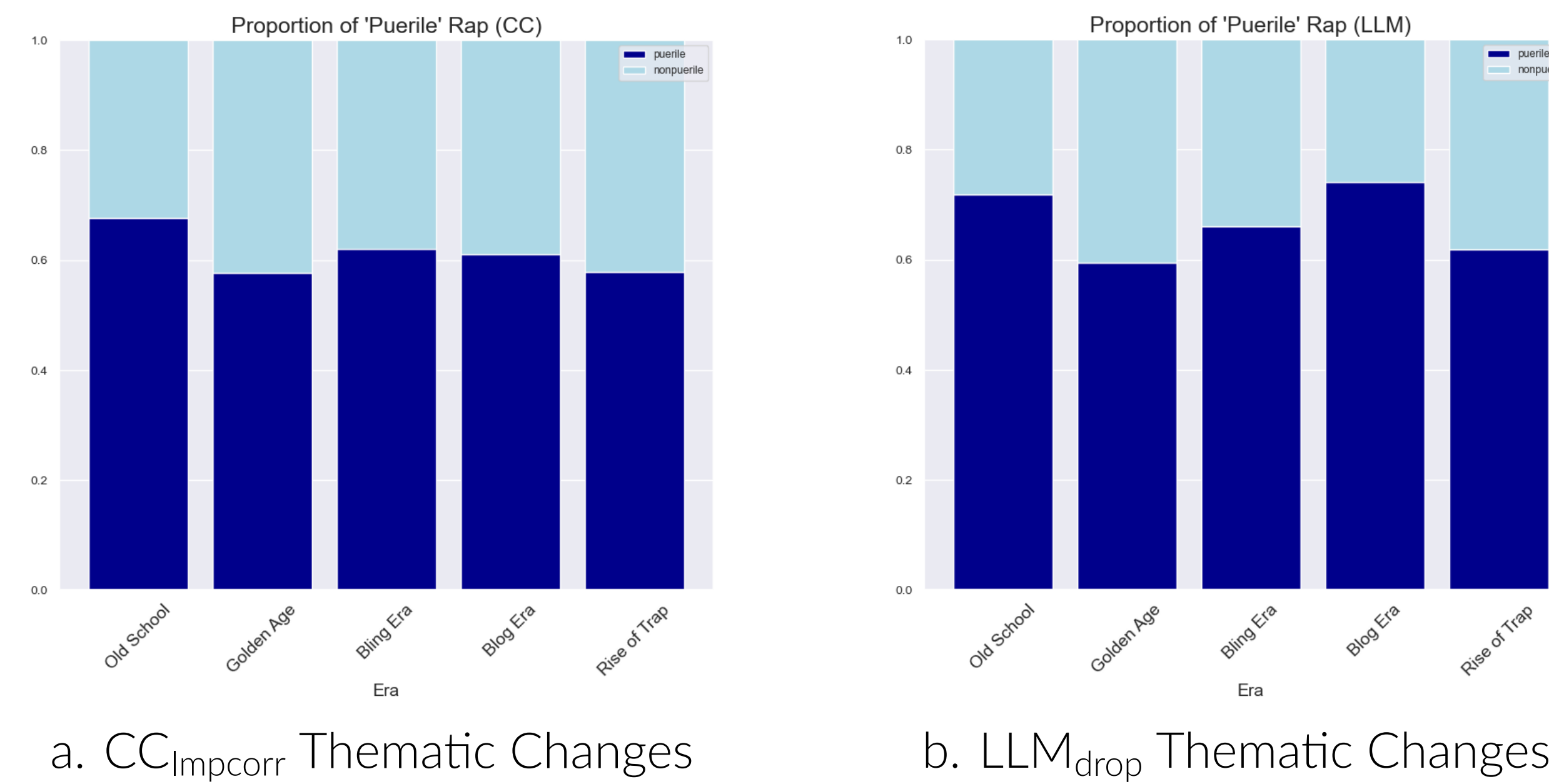


Figure 2. Yearly changes in theme, according to the top-performing models. The CC model suggests an increase in the proportion of braggadocious songs from the Bling Era, aligning with prior research.

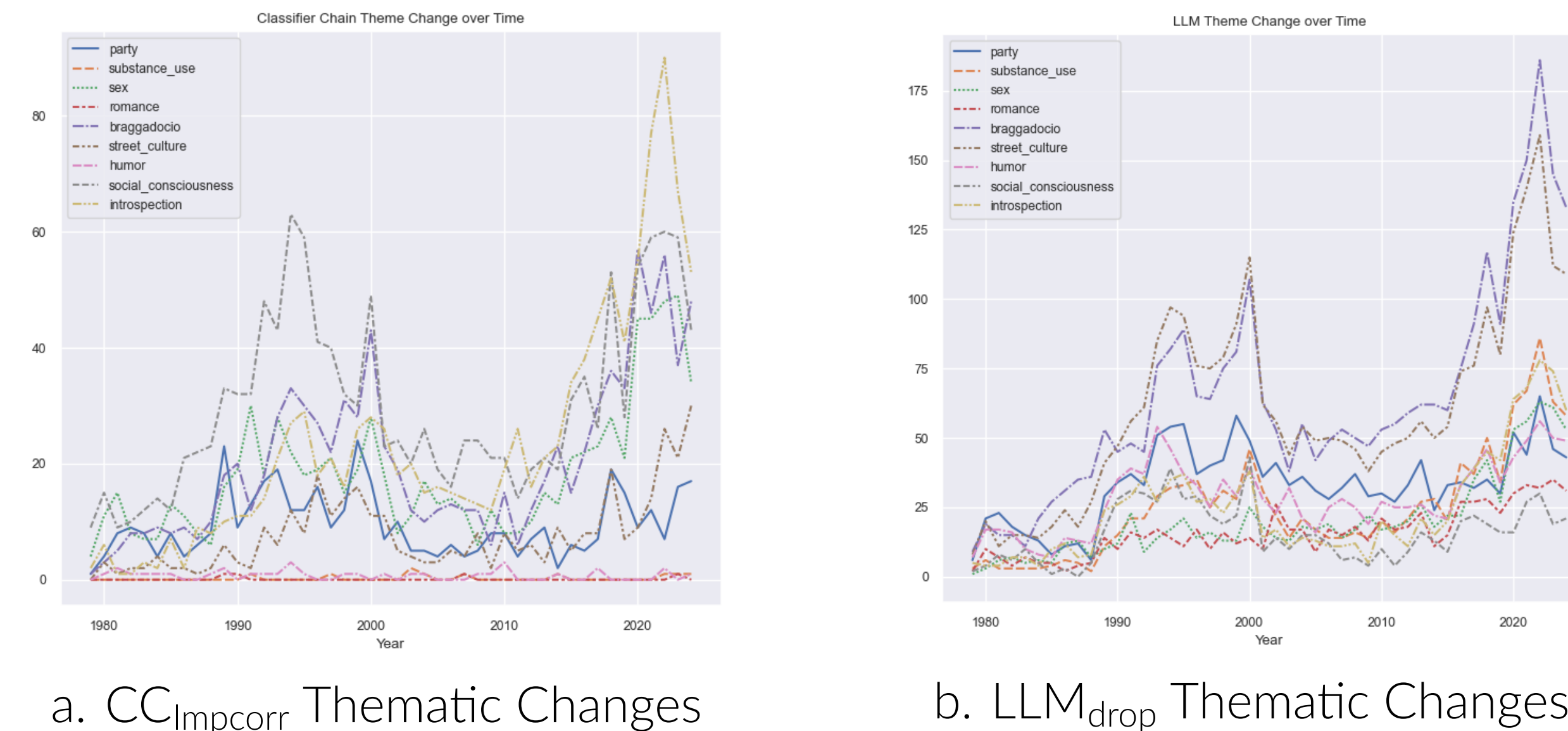


Figure 3. Distribution of Themes based on CC_{impcorr}. According to this method, rap lyrics are mostly about partying and bragging.

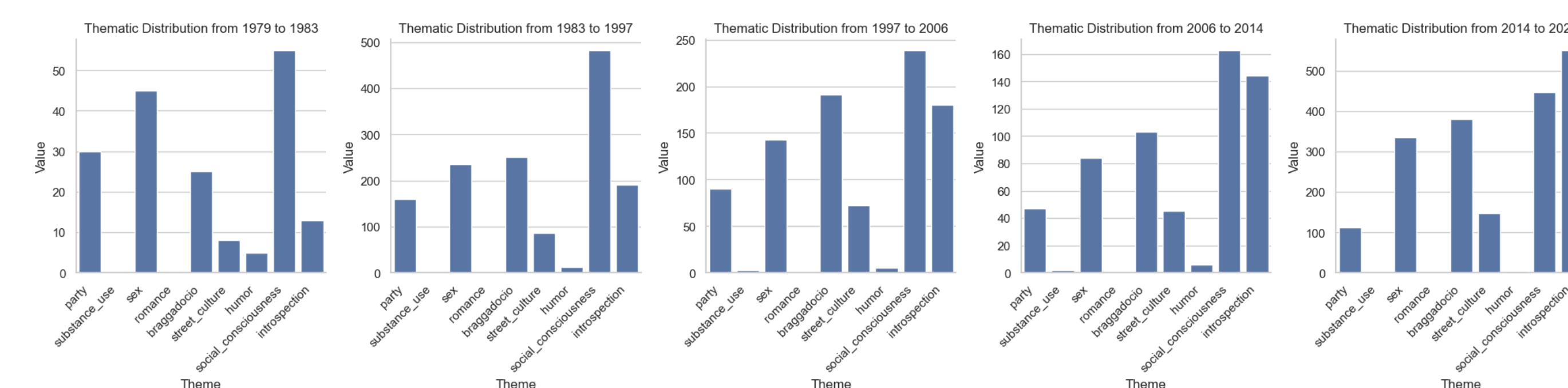
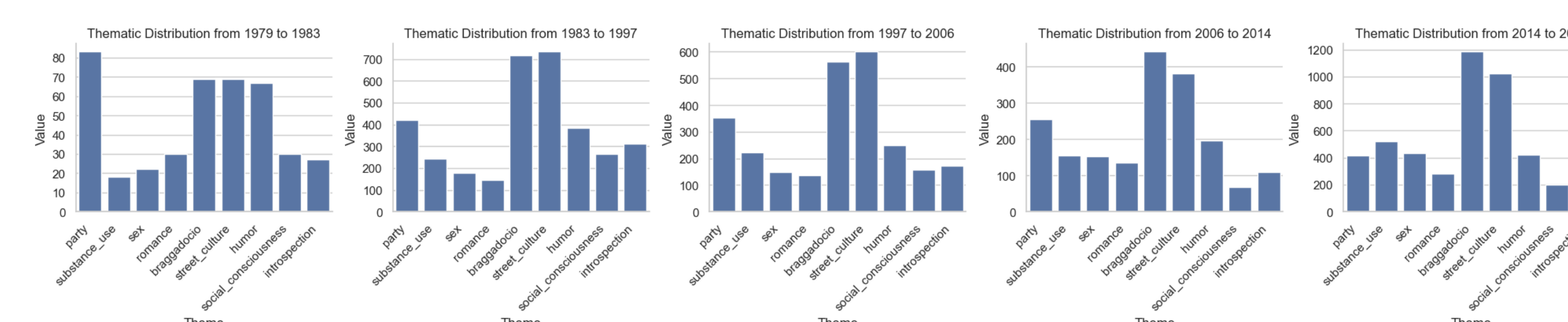


Figure 4. Distribution of Themes based on LLM_{drop}. According to this model, rap lyrics are mostly about bragging and street culture.



Model Performance

Table 1. Performance of Each Algorithm for Each Evaluation Metric

Metric	BR	LP	CC _{Ensemble}	CC _{impcorr}	MIKNN	LLM	LLM _{drop}
Hamming Loss	0.198 (1)	0.274 (5)	0.216 (3)	0.211 (2)	0.224 (4)	0.312 (7)	0.296 (6)
Accuracy	0.005 (7)	0.024 (5)	0.033 (3)	0.029 (4)	0.019 (6)	0.047 (2)	0.058 (1)
Micro F1	0.005 (7)	0.137 (3)	0.104 (4)	0.090 (6)	0.092 (5)	0.421 (2)	0.425 (1)
Macro F1	0.004 (7)	0.076 (3)	0.065 (4)	0.058 (5)	0.042 (6)	0.423 (1)	0.420 (2)
Precision	0.100 (7)	0.186 (6)	0.272 (4)	0.278 (3)	0.218 (5)	0.311 (2)	0.319 (1)
Recall	0.003 (7)	0.109 (3)	0.065 (4)	0.054 (6)	0.062 (5)	0.649 (1)	0.636 (2)
Average Rank	6	4.17	3.67	4.33	5.17	2.5	2.17

Model Selection

Classifier Chain

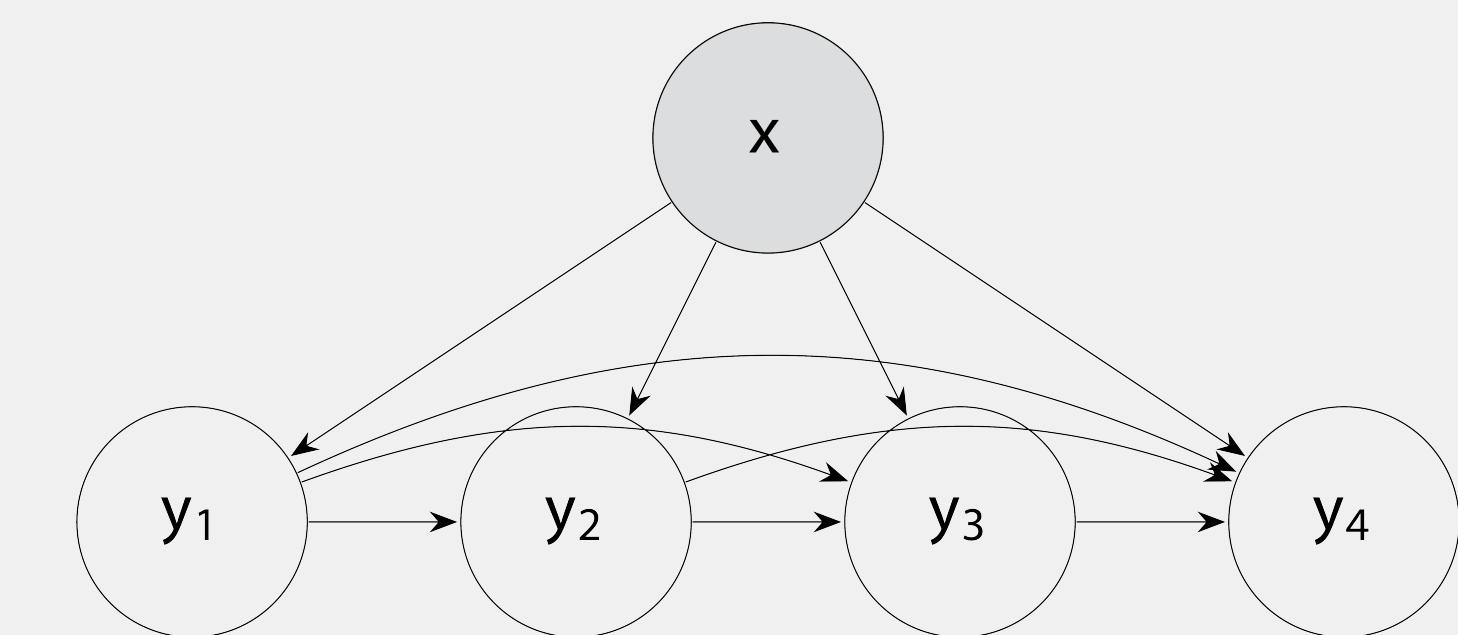


Figure 5. Modeling the relationships between labels in a classifier chain.

Classifier chains turn the multilabel classification problem into a series of binary relevance problems, but models the relationship between labels. To decide on the optimal label ordering, this research adapted the Imprecise Correlation (**Impcorr**) method. Given the distribution of labels in the training data, this algorithm calculates

$$ISU(y_j, y_k) = \frac{2IIG(y_j, y_k)}{S^*(P(y_j)) + S^*(P(y_k))}$$

and finds the next most likely label.

Large Language Model

LLaMa 3.2 is an open source large language model. To adjust the model for these research purposes, I used Few-Shot and Chain-of-Thought prompting. The temperature of the model was 0.75 and the system message is available on the GitHub for this project.

References

- [1] Michael Eric Dyson. *Know What I Mean? : Reflections on Hip-Hop*. Basic Books, New York, UNITED STATES, 2007.
- [2] Jennifer C. Lena. Social context and musical content of rap music, 1979-1995. *Social Forces*, 85(1):479–495, 2006.
- [3] Serafin. Moral-García, Javier G. Castellano, Carlos J. Mantas, and Joaquín Abellán. A new label ordering method in classifier chains based on imprecise probabilities. *Neurocomputing*, 487:34–45, 2022.
- [4] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. Llama: Open and efficient foundation language models, 2023.